

PRANAV SWAROOP GUNDLA

Doctoral Researcher in Computational Oncology

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EDUCATION

University of Duisburg-Essen / University Hospital Essen

Doctoral Researcher, Computational Oncology — West German Cancer Center and IKIM

Kocakavuk Lab; CANTAR Network. Research focus: multimodal omics, cancer ecotypes, computational pathology, and molecular prediction in diffuse gliomas.

Essen, Germany

July 2023–Present

Manipal Academy of Higher Education

M.Sc. Bioinformatics

Manipal, India

2018–2020

Université Grenoble Alpes

International M2 Track, Health Engineering — Healthy Living Technologies

Research internship: Institute for Advanced Biosciences, SigNet Group; TP53-associated lung adenocarcinoma cartography.

Grenoble, France

2019–2020

Osmania University

B.Sc. Mathematics, Electronics, and Computer Science

Hyderabad, India

2015–2018

RESEARCH EXPERTISE

Computational pathology; whole-slide imaging; adult diffuse gliomas; molecular subtype prediction; weakly supervised learning; attention-based multiple-instance learning; vision transformers; pathology foundation models; multimodal and spatial omics; tumor microenvironment; interpretable AI; cross-cohort validation.

COMPUTATIONAL SKILLS AND TECHNIQUES

Machine learning: PyTorch, TensorFlow, scikit-learn, CNNs, vision transformers, foundation models, MIL, attention models

Computational pathology: WSI processing, H&E patching, OpenSlide, QuPath, OpenCV, feature extraction, attention heatmaps, stain normalization

Bioinformatics and omics: transcriptomics, genomics, NGS, multi-omics integration, spatial omics, Scanpy, AnnData, differential expression, enrichment and survival analysis

Programming and data: Python, R, Julia, Bash, JavaScript/TypeScript, NumPy, pandas, Jupyter

Research computing: SLURM HPC, Docker, Apptainer/Singularity, Snakemake, Nextflow, Git/GitHub, GCP, AWS, Azure

Visualization: R Shiny, Plotly, D3

RESEARCH EXPERIENCE

West German Cancer Center / IKIM, University Hospital Essen

Doctoral Researcher, Kocakavuk Lab

Essen, Germany

July 2023–Present

- Develop deep-learning models for molecular subtype prediction from glioma H&E whole-slide images.
- Train weakly supervised and attention-based MIL models across approximately 5,000 public whole-slide images.
- Build end-to-end workflows spanning patching, feature extraction, aggregation, molecular prediction, validation, and explainability.
- Evaluate IDH mutation, 1p/19q codeletion, and CDKN2A/B alteration prediction; connect attention-defined morphology with genotype.
- Integrate imaging, transcriptomic, genomic, and spatial context to study cancer ecotypes and tumor microenvironment.
- Run reproducible external-cohort validation using SLURM, Docker/Apptainer, and version-controlled workflows.

Institute for Advanced Biosciences, SigNet Group

M.Sc. Research Intern

Grenoble, France

January–June 2020

- Integrated omics and signaling-flux analyses for lung adenocarcinoma biomarker and therapeutic-target discovery.
- Completed thesis research on TP53-associated lung adenocarcinoma cartography.

PUBLICATIONS AND CONTRIBUTIONS

Gundla, P. S., Atak, F., Nezamabadi, M., et al. (2025). 278P Genetic subtype prediction in diffuse gliomas with a vision transformer-based model. ESMO Real World Data and Digital Oncology, 10, 100474. [DOI](#)

Contribution: First author; led WSI preprocessing, vision-transformer/MIL pipeline development, molecular-subtype modeling, external validation, attention-based interpretation, and abstract/poster preparation.

Alotaibi, A., Gadekar, V. P., Gundla, P. S., et al. (2023). A comprehensive analysis of mRNA expression profiles of esophageal squamous cell carcinoma reveals downregulation of Desmoglein 1 and crucial genomic targets. Cancer Biomarkers, 38(4), 465–487. [DOI](#)

Contribution: Contributed to transcriptomic meta-analysis, candidate-biomarker identification, pathway and survival analysis, and biological interpretation.

Alotaibi, A., Gadekar, V. P., Gundla, P. S., et al. (2023). Global comparative transcriptomes uncover novel and population-specific gene expression in esophageal squamous cell carcinoma. Infectious Agents and Cancer, 18, 47. [DOI](#)

Contribution: Contributed to comparative transcriptome analysis, differential-expression and biomarker discovery, enrichment analysis, and interpretation across populations.

Chowdhury, T., Tesileanu, C. M. S., Kocakavuk, E., et al.; GLASS Consortium. (2025). Tumor-initiating genetics and therapy drive divergent molecular evolution in IDH-mutant gliomas. *bioRxiv*. [Preprint] [DOI](#)
Contribution: GLASS Consortium co-author; contributed to collaborative analysis and interpretation of longitudinal IDH-mutant glioma evolution.

Gundla, P. S., et al. (2026). Genetic subtype prediction in diffuse gliomas with a vision transformer-based model. [Manuscript under revision]
Contribution: First author; designed the algorithm and pipeline, evaluated the model, developed attention-based interpretation and longitudinal analyses, and prepared and reviewed the manuscript.

SELECTED SOFTWARE AND RESEARCH PROJECTS

[Histo2GeneCLAM](#) — research implementation for predicting immune and inflammatory gene signatures from whole-slide images using patch-based models, classic MIL, and CLAM, with external validation, stain normalization, and attention visualization.

[Pathopatcher](#) — contributed a targeted workflow fix correcting output generation.

[RNA-Seq-Automation](#) — Bash automation for RNA-seq analysis using new-Tuxedo workflows.

[SplitSeq-nf](#) — Nextflow workflow for splitting FASTA sequences with NCBI BLAST+ tooling.

[Scientific portfolio](#) — JavaScript site communicating computational-pathology research, methods, publications, and scientific progress.

GRANTS, AWARDS, AND HONORS

ESMO Merit Award, 1st ESMO AI Congress — poster 278P 2025

First Place, Hackathon, Congrès National des Pharmaciens, Bordeaux 2020

IDEX & UGA Foundation Scholarship, Université Grenoble Alpes 2019

CONFERENCE PRESENTATIONS AND INVITED TALKS

“Genetic subtype prediction in diffuse gliomas with a vision transformer-based model,” poster 278P, 1st ESMO AI Congress 2025

Invited talk, CANcer TARgeting PhD/PostDoc Retreat, CIO Cologne

ESCC mRNA-signature presentation, 46th Annual Meeting of Indian Society of Human Genetics, Manipal April 2022

TEACHING, MENTORING, AND RESEARCH SERVICE

Mentor, Anastasija Ivanacuka — glioma classification with MGMT methylation; introductory computational training 2026

Mentor, Fatma Atak — computational oncology, RNA-seq, glioma classification, and AI in medicine 2025

Mentor, two M.Sc. students in computational research projects

Lead, laboratory journal clubs, technical discussions, and internal workshops

Teaching assistant, command-line genomics and R programming 2019

Private mathematics tutor 2021

RELATED PROFESSIONAL EXPERIENCE

Plantik Biosciences Paris, France
Associate Bioinformatician November 2020–May 2023

- Developed bioinformatics solutions for genomics and plant-breeding research.
- Built Python/R workflows and cloud infrastructure using GCP and AWS.
- Integrated biological data for hypothesis generation, target identification, and prioritization.

Mbiomics LLC Remote, United States
Bioinformatics Analyst, Freelance February–December 2022

- Performed esophageal squamous cell carcinoma transcriptomic analysis and biomarker discovery.
- Contributed analyses and interpretation supporting two peer-reviewed publications.

SELECTED TRAINING AND CERTIFICATIONS

Precision Oncology Summer School, European Scientific Institute 2020

Biohealth Computing Summer School, European Scientific Institute 2019

Algorithms for DNA Sequencing, Johns Hopkins University

Command Line Tools for Genomic Data Science, Johns Hopkins University

Supervised Machine Learning: Regression & Classification, DeepLearning.AI / Stanford Online

Python for Data Science and AI, IBM

Introduction to Julia, JuliaAcademy

PROFESSIONAL AFFILIATIONS

European Society for Medical Oncology (ESMO), Member (ID 701003) | Society for Neuro-Oncology | European Association for Cancer Research

LANGUAGES

English (full professional proficiency); French and German (A1); Telugu (native); Hindi and Tamil; Kannada (reading and comprehension).